ABEE RAVENAA M

2027

QUANTUM SOFTWARE ENGINEERING INTERN CANDIDATE

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Professional Summary

Final-year Electronics and Communication Engineering undergraduate with strong foundations in Quantum Computing, Software Development, Artificial Intelligence, and Machine Learning. Proficient in Python, SQL, GitHub, Docker, and Qiskit, with experience in developing simulation and optimization platforms using hybrid quantum-classical approaches. Worked on Quantum Chip Design and Architecture Simulation and Quantum Data Filtering and Feature Selection platforms, applying Quantum Machine Learning, Data Analysis, Optimization, and Software Engineering principles. Seeking internship opportunities in Quantum Software Engineering, AI/ML, Software Development, and Research-oriented technology roles.

Education

EXCEL ENGINEERING COLLEGE (AUTONOMOUS) — *Tamil Nadu, India*

B.E. - ELECTRONICS AND COMMUNICATION ENGINEERING — *Expected Graduation: 2027 | CGPA: 8.8/10*

Technical Skills

Programming	Python, SQL
Software Development	Flask, REST APIs, Software Engineering
Quantum Computing	Qiskit, Quantum Circuits, Quantum Gates, Quantum Algorithms, Quantum Machine Learning, Variational Quantum Circuits, Quantum Simulation, Hybrid Quantum-Classical Computing
Artificial Intelligence	Machine Learning, Deep Learning, Data Analysis, Feature Engineering, Scikit-Learn
Developer Tools	Git, GitHub, Docker, VS Code
Libraries	NumPy, Pandas, Matplotlib
ECE Fundamentals	Digital Electronics, Embedded Systems, Signal Processing, Semiconductor Devices, VLSI Fundamentals
Core Concepts	Data Structures, Algorithms, Object-Oriented Programming, Problem Solving

Projects

Quantum Chip Design and Architecture Simulator

PYTHON, QISKIT, NETWORKX, FLASK

- Developed a software-based simulator for designing and visualizing quantum chip architectures and qubit connectivity.
- Implemented quantum circuit mapping, architecture analysis, and connectivity optimization using graph-based approaches.
- Designed interactive visualizations to evaluate routing efficiency, scalability, and quantum hardware constraints.

Quantum Data Filtering and Feature Selection Platform

PYTHON, PANDAS, SCIKIT-LEARN, QISKIT

- Developed a hybrid quantum-classical platform for filtering noisy datasets and identifying relevant features for machine learning models.
- Implemented data preprocessing, feature engineering, and quantum-enhanced optimization workflows.
- Compared classical and quantum-inspired feature selection techniques to improve model efficiency and predictive performance.

Certifications

IBM Quantum Learning, Basics of Quantum Information

IBM Quantum Learning, Quantum Business Foundations

IBM Qiskit Learning Path, Introduction to Quantum Computing and Quantum Hardware

DeepLearning.AI, Machine Learning Specialization

Achievements

, Developed software-based quantum simulation and optimization platforms using Python and Qiskit

, Designed quantum-inspired solutions for chip architecture analysis and machine learning feature selection

, Completed industry-recognized certifications in Quantum Computing and Artificial Intelligence

, Participated in technical workshops, webinars, and self-learning programs focused on Quantum Technologies, AI, and Software Development